Bonus Guide 3 of 5

THE POTTERY TEMPLATE VAULT

Engobes, Pigments & Oxides Guide

Color with engobes, underglazes, oxides & commercial pigments

Engobes, Pigments & Oxides Guide

> Before glaze — and sometimes instead of it — potters use engobes, underglazes, oxides and commercial pigments to add color, depth and surface interest. This guide explains each one and when to use it.

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1 · The Difference: Engobe, Underglaze, Oxide, Stain

Material	What it is	When applied	Used with glaze?
Engobe / colored slip	Liquid clay with added colorant, matched to body shrinkage	Plastic or leather-hard clay	Yes (or unglazed)
Underglaze	Color suspended in a clay-flux base, low shrinkage	Plastic clay through bisqueware	Always — needs a clear topcoat
Metal oxide	Pure or mixed metal pigment (iron, cobalt, copper, manganese, etc.)	Bisqueware (usually)	Yes — can be mixed into glaze or applied under/over
Commercial stain	Pre-fritted stable pigment, wide color range, food-safe	Mixed into slips, engobes, glazes or clay body	Flexible

2 · Engobes — Colored Slips

An engobe is a slip (liquid clay) with added colorant. Because it contains clay, it must have a **shrinkage rate close to your clay body**. Applied to plastic or soft leather-hard clay, it dries with the

piece.

Making a simple engobe

1. Start with a base slip recipe: 50% ball clay + 50% kaolin, mixed to the consistency of single cream.
2. Add 5–15% metal oxide or commercial stain by dry weight to color it.
3. Sieve through a 100-mesh screen.
4. Test on a tile matched to your clay body — fire and check for peeling or flaking.

Applying engobe

- **Brush** — 2–3 coats on leather-hard clay
- **Dip** — for overall coverage on plastic clay
- **Pour** — for the interior of a vessel
- **Sgraffito** — scratch through dried engobe to reveal the clay color beneath

> ⚠️ **Warning**
> Never apply engobe to bone-dry clay — it will crack and peel. The clay underneath must still have enough moisture to accept the extra water in the slip.

3 · Underglazes

Underglazes are the most versatile colorant for detailed painted decoration. They remain stable at high temperatures and do not run. Because they contain little or no flux, they must always be fired **under a clear glaze topcoat** to produce their full color and a food-safe finish.

Working with underglazes

- Apply to bisqueware for the sharpest lines and most control.
- Can also be applied to leather-hard or plastic clay, but lines may blur slightly as the clay shrinks.
- Thin with water for watercolor-style washes; use undiluted for solid coverage.
- 2–3 coats give full opacity; 1 coat gives a translucent tint.
- Let each coat dry before applying the next.
- Apply a clear or white satin glaze on top before the final firing.

> 💡 **Tip**
> You can fire underglazes to bisque temperature (1000 °C) before applying the topcoat glaze. This "fixes" the design and makes the final glaze application easier without disturbing your painted work.

4 · Metal Oxides

Metal oxides are the oldest ceramic colorants. They are raw, reactive and can produce complex, variable effects depending on firing temperature and atmosphere.

Oxide	Color (oxidation)	Color (reduction)	Typical percentage
Iron oxide (red)	Yellow to amber, brown, black at high %	Celadon green to grey	2–15%
Cobalt carbonate	Blue — powerful, use sparingly	Blue (similar)	0.5–2%
Copper carbonate	Green to turquoise	Copper red	1–4%
Manganese dioxide	Brown, purple, black	Similar	2–6%
Chrome oxide	Olive green	Dark green	1–3%
Rutile (titanium + iron)	Cream, tan, mottled	Mottled, crystalline	3–8%
Tin oxide	White, opacifier	White	5–10%

Oxide wash technique

1. Mix oxide in water to a thin, milky consistency (1–5% oxide by weight).
2. Brush over bisqueware. The wash settles into recessed areas — carved lines, texture, joints.
3. Wipe the raised surfaces lightly with a damp sponge while still wet, leaving the oxide in the low points.
4. Apply a clear or celadon glaze over the top.
5. The result: a natural-looking shading that follows the form.

> ⚠️ **Warning**
> Manganese is a respiratory hazard in dry form. Always wear a respirator when handling dry oxide powders. Wash hands and work surfaces after use. Once fired, the piece is safe.

5 · Commercial Stains and Body Stains

Commercial ceramic stains are pre-fritted (pre-processed) pigments that are temperature-stable and food-safe. They give a wider, more predictable range of colors than raw oxides.

Uses

- **Mixed into a glaze** — typically 5–15% stain by dry weight
- **Mixed into a slip or engobe** — 5–20%
- **Mixed into the clay body** — 5–10% for a colored clay body throughout
- **Applied directly** — dry stain in a thin gum solution, brushed on bisqueware under a clear glaze

⭐ **Important**

Stains interact differently with different glaze bases. Always test a new stain in your specific glaze before committing to a production run. Some stains are incompatible with certain glaze bases (e.g., chrome stains and zinc-containing glazes).

6 · Application Techniques

Painterly approach (underglaze on bisque)

Treat the bisqueware surface like watercolor paper. Build up colors in layers from light to dark. Use fine brushes for detail, wide flat brushes for backgrounds.

Oxide wash over carved work

Apply a thin iron or cobalt wash over carved or stamped bisqueware, then wipe the high points. The oxide pools in the recesses, creating instant depth and contrast.

Wax resist with multiple colors

1. Apply underglaze color 1, let dry.
2. Wax over the areas of color 1 you want to keep.
3. Apply underglaze color 2 over everything — it sticks only where there is no wax.
4. Fire. The wax burns away; both colors survive under the topcoat glaze.

7 · Combining with Glazes

Combination	Effect	Tips
Underglaze + clear gloss	Bright, saturated colors; food-safe	Most common for painted work
Underglaze + white satin	Slightly muted, soft finish	Better for pastels and light tones
Oxide wash + celadon	Earthy, classical Chinese-influenced look	Works especially well on carved pieces
Engobe + unglazed (bisque fired)	Matte, earthy, rough texture visible	Not food-safe for inside of food vessels; fine for decorative work
Stain in glaze + clear topcoat	Even, predictable color through the glaze	Test percentages — stains vary in tinting strength

8 · Color Chart Reference (oxidation, cone 6)

Colorant	1%	3%	5%	8%
Iron oxide (red)	Pale yellow	Warm amber	Brown-orange	Dark brown
Cobalt carbonate	Light blue	Medium blue	Deep blue	Very dark blue
Copper carbonate	Pale green	Medium green	Deep green	Dark green / black
Manganese dioxide	Pale mauve	Brown-purple	Dark purple	Near black

These are indicative only — results vary with glaze base, firing temperature, kiln atmosphere and clay body. Always test.

9 · Troubleshooting

Problem	Cause	Fix
Engobe peels after drying	Applied to bone-dry clay; shrinkage mismatch	Apply to leather-hard; reformulate to match clay body
Underglaze color washes out after firing	Too thin; topcoat glaze too thick or too fluid	Apply 3 coats underglaze; use a satin clear rather than a fluid gloss
Oxide color bleeds into adjacent glaze	Oxide applied too thick; glaze too fluid	Thin the oxide wash; use a stiffer glaze
Color completely different from expected	Kiln atmosphere; wrong temperature cone	Verify cone with witness cones; test in oxidation and reduction separately

Before you apply any colorant

- Is the clay/bisque surface clean, dry and dust-free?
- Have I tested this colorant on a tile with my specific clay body and glaze?
- Do I know what topcoat (if any) goes over it?
- Is the colorant at the right consistency for the technique?

Next: Glaze Layering Guide — combining multiple glazes for depth, texture and unique effects.